Multiple-Impulse Spacecraft Rendezvous Optimization

Hasnain

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Abstract

Writing prompt: Write this last. In one paragraph, state the rendezvous problem, the model and search method used, the main quantitative result, and the most important conclusion or limitation. Leave results unspecified until you have verified them.

1 Introduction and Aim

1.1 Spacecraft Trajectory Optimization Problem

This problem discusses how a spacecraft in a distnct orbit from another object in an orbit can rendezvous at a specific location in space at a particular time with the same velocity while minimizing the amount of fuel usage. For this specific project, we will be focusing on 2 spacecrafts in orbit around Earth where one is defined as the *chaser* spacecraft and the other is the *target* spacecraft.

1.2 Project Aim and Research Question

This leads us to the question that I aim to work around:

How can a spacecraft use a small number of engine burns to rendezvous with another spacecraft orbiting Earth?

Thus, the aim of this project that follows is to develop a simulator for orbital rendezvous missions. I will begin with a simplified model in 2 dimenions and only considering the Earth as the only relevant gravitational body. I will look to minimize the following 2 errors:

Position Error:

$$J_r = |r_c(T) - r_t(T)|^2 \tag{1}$$

Velocity Error:

$$J_v = |v_c(T) - v_t(T)|^2 \tag{2}$$

In addition to position and velocity error, I will be taking into account fuel usage with the assumption that fuel usage is directly proportional to the change in velocity in the x, y, and z directions. These three will be combined into the following cost function:

$$J = \omega_r J_r + \omega_v J_v + \omega_{\Delta v} \sum_i |\Delta v_i| \tag{3}$$

After testing with the 2 dimensional model, the aim will be to expand to 3 dimensions and will consider the Moon, Sun and Jupiter as gravitationally relevant bodies to the motion of the spacecrafts.

2 Physical Model and Problem Definition

2.1 Assumptions and equations of motion

To begin, the following assumptions will be used for the 2 dimensional model

- Planar motion in an Earth-centred inertial Cartesian frame.
- Central gravity with constant gravitational parameter μ.
- An uncontrolled target and a chaser with instantaneous burns.
- Fuel usage is directly proportional to change in velocity of a burn.

For each spacecraft, position is defined as $\mathbf{r} = \begin{bmatrix} x \\ y \end{bmatrix}$ and velocity is defined as $\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$:

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\mu \frac{\mathbf{r}}{\|\mathbf{r}\|^3}. \tag{4}$$

Equation (4) defines the ODE that the spacecraft states will be modelled after, with μ defined as the gravitational parameter and for the 2 dimensional model is the gravitational parameter of Earth:

$$\mu = GM$$
$$\mu = 3.986 \times 10^{14}$$

At burn time t_i, position is continuous and velocity changes according to

$$\mathbf{v}(t_i^+) = \mathbf{v}(t_i^-) + \Delta\mathbf{v}_i. \tag{5}$$

Equation (5) is the mathematical representation of the assumption that all burns are applied as instantaneous burns, meaning that position does not change but the velocity changes instantly right after time t_i.

For the three-dimensional model, the assumptions are extended as follows:

- Three-dimensional motion in a Sun-centred Cartesian frame, treated as inertial under the approximation that the Sun is physically fixed at the origin with zero velocity.
- Newtonian point-mass gravity from the Sun, Earth, Moon, and Jupiter, each with a constant gravitational parameter μ_b. Earth, the Moon and Jupiter move under the gravity of all four bodies. The Sun attracts the moving bodies but does not respond to their gravitational attraction. Spacecraft masses are negligible, and other gravitational bodies and non-gravitational forces are neglected.
- An uncontrolled target and a chaser with instantaneous burns. Position remains continuous across each burn, while velocity changes instantaneously.
- Total impulse magnitude, $\sum_i \|\Delta\mathbf{v}_i\|$, is used as a proxy for propellant consumption and is part of the third term for Equation 3, the cost function.

2.2 Scenario and definition of success

We can begin with defining the initial state of a spacecraft using the position and velocity vectors in 2 dimensions as follows:

$$\tilde{\mathbf{v}} = \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix} \tag{6}$$

Then applying ODE in Equation 4 to the derivative of the initial state, we obtain:

$$\tilde{\mathbf{v}'} = \begin{bmatrix} v_x \\ v_y \\ a_x \\ a_y \end{bmatrix} \tag{7}$$

$$\tilde{\mathbf{v}'} = \begin{bmatrix} v_x \\ v_y \\ -\mu \frac{x}{|r|^3} \\ -\mu \frac{y}{|r|^3} \end{bmatrix} \tag{8}$$

Then with any final time T, the number of burns and the initial state of both spacecrafts defined as shown in Equation 6 the rendezvous between the *chaser* and *target* spacecraft can be determined.

For the initial state of the 3 dimensional model with more than one significant gravitational body, we can define the acceleration using the same base ODE from Equation 4, but make the following changes to the original model:

$$\ddot{\mathbf{r}} = -\mu \frac{\mathbf{r}}{\|\mathbf{r}\|^3}.$$

$$\ddot{\mathbf{r}} = \sum_{i=0}^{n-1} -\mu_i \frac{r_i}{|r_i|^3} \tag{9}$$

Here, n is the number of gravitational bodies being considered, and where μ_i and r_i correspond to the i-th gravitational body on a 0 index. For my case, the gravitational bodies will be defined as follows:

$$[\text{Sun}, \text{Earth}, \text{Moon}, \text{Jupiter}]$$

The following gravitational parameters, μ will be used for the gravitational bodies:

Body	μ ($\text{m}^3\,\text{s}^{-2}$)
Sun	1.3271×10^{20}
Earth	3.986×10^{14}
Moon	4.9028×10^{12}
Jupiter	1.2671×10^{17}

Furthermore, we will define a successful rendezvous as follows:

- Final Separation $<= 1000m$
- Velocity Difference $<= 1ms^{-1}$

Final docking and approach are not within the scope of the project so this is sufficient for a successful rendezvous

3 Computational Method

3.1 Trajectory propagation

To begin, I use Python to calculate how the initial states of both spacecrafts change over time. The equations of motion are solved using the `solve_ivp` function from SciPy, with the `DOP853` method.

This method adjusts the size of its time steps as the trajectory is calculated. For both models, I set the relative tolerance to 10^{-6} and leave the absolute tolerance at its default value of 10^{-6}. These tolerances control the numerical integration error; they do not directly define the final separation required for a successful rendezvous.

3.1.1 Two-dimensional model

For the original 2 dimensional model, the initial states are entered in the form shown in Equation 6, with positions in metres and velocities in metres per second relative to the fixed Earth. The final time T and number of burns are also specified before the search begins. The target is first propagated from $t = 0$ to $t = T$ without any burns. I keep a continuous interpolated solution of its state using `dense_output`, allowing its position and velocity to be obtained at any time within the simulation. This target solution is reused when testing different burn sequences, since the chaser's burns do not affect the target's motion.

For the chaser, the trajectory is divided into intervals separated by the burn times. The ODE is first solved from the initial time to the first burn. At that time, the change in velocity is added to the state while position remains unchanged, as described in Equation 5. The updated state is then used as the initial state for the next interval. This process is repeated for each burn, followed by a final propagation to T. Thus, both spacecrafts are compared at the same final time even though the chaser's motion is calculated in separate intervals.

3.1.2 Three-dimensional model with multiple gravitational bodies

For the 3 dimensional model, the initial spacecraft states contain six values, $[x, y, z, v_x, v_y, v_z]$. I first obtain the initial states of Earth, the Moon and Jupiter from JPL Horizons, using the default starting epoch of 27 October 2026 at 12:00:00 UTC. These values are converted from kilometres and kilometres per second to SI units. The Sun is fixed at the origin with zero velocity throughout the simulation.

Earth, the Moon and Jupiter are then propagated under their mutual gravitational attraction and the Sun's gravity. I keep the same combined state of 24 values, with six values for each body, but set all six derivatives of the Sun's state to zero. This calculation is performed once before the rendezvous search, using the same solver and tolerances as the spacecraft propagation. A continuous interpolated solution is retained so that the location of each body can be obtained at the times requested by the spacecraft solver. I chose to hold the Sun fixed to simplify the model by neglecting its response to the other bodies.

The spacecraft states entered into the simulator describe their initial positions and velocities relative to Earth. I add Earth's initial position and velocity so that the spacecrafts and gravitational bodies can be tracked together with the Sun at the origin. At each time, I calculate the distance and direction from each body to the spacecraft and add the four gravitational contributions to obtain its acceleration:

$$\mathbf{a}(t) = -\sum_{b=1}^{4} \mu_b \frac{\mathbf{r}(t) - \mathbf{r}_b(t)}{\|\mathbf{r}(t) - \mathbf{r}_b(t)\|^3}, \tag{10}$$

where $\mathbf{r}(t)$ and $\mathbf{r}_b(t)$ are the positions of the spacecraft and gravitational body b, respectively, measured from the Sun. Their difference gives the spacecraft's displacement from that body. The target is propagated once for the search, and the chaser is propagated between burns in the same way as the 2 dimensional model. Both spacecrafts use the same precomputed body trajectories. Their masses are neglected, so neither spacecraft changes the motion of the four bodies or the other spacecraft.

3.2 Burn representation

In both implementations, a burn is stored as a pair (d_i, t_i), where d_i is a signed burn magnitude and t_i is the time of the burn. To simplify the search, burns are restricted to act along or opposite to the chaser's motion relative to Earth just before the burn. The resulting velocity change is added to the chaser's velocity, while its position remains unchanged. This is expressed as follows:

$$\Delta \mathbf{v}_i = d_i \frac{\mathbf{v}_{c/E}(t_i^-)}{\|\mathbf{v}_{c/E}(t_i^-)\|}. \tag{11}$$

A positive value of d_i applies the burn along this velocity, while a negative value applies it in the opposite direction. Thus, each burn has two search variables rather than separate velocity changes in each coordinate direction.

For the 2 dimensional model, Earth is fixed, so $\mathbf{v}_{c/E} = \mathbf{v}_c$. For the 3 dimensional model, I obtain Earth's velocity from the precomputed body solution at each burn time and subtract it from the chaser's velocity:

$$\mathbf{v}_{c/E}(t_i^-) = \mathbf{v}_c(t_i^-) - \mathbf{v}_E(t_i). \tag{12}$$

This gives the direction of the spacecraft's motion relative to Earth at the time of the burn. The same calculation is used when scoring candidate sequences and when reconstructing the selected trajectory for plotting.

3.3 Objective function and search

After each candidate burn sequence has been propagated, I calculate the squared position and velocity differences at T. The total change in velocity is calculated by adding the absolute values of the signed burn magnitudes:

$$\Delta v_{\text{total}} = \sum_{i=1}^{N} |d_i| = \sum_{i=1}^{N} \|\Delta \mathbf{v}_i\|. \tag{13}$$

These three quantities are combined using the cost function defined in Equation (3) to score each candidate burn sequence. The weights determine how much each term contributes to the search. This scaling is necessary because position error, velocity error and total change in velocity have different units and numerical sizes. For the original 2 dimensional script, the weights are fixed as

$$(\omega_r, \omega_v, \omega_{\Delta v}) = (10^{-8}, 10^{-2}, 10^{-3}), \tag{14}$$

when the states and burns are expressed in SI units. In the 3 dimensional script, the three positive weights are supplied when running the program, allowing different choices to be tested.

To search for a burn sequence, I use SciPy's `differential_evolution` function. This method starts with a population of candidate sequences, generates new candidates by combining and changing their parameter values, and keeps candidates with lower costs. For N burns, there are $2N$ parameters to search. The burn magnitude limits are 500 $\mathrm{m\,s^{-1}}$ for the original 2 dimensional model and 1000 $\mathrm{m\,s^{-1}}$ for the 3 dimensional model. Thus, for the 3 dimensional search, the bounds are

$$-1000\ \mathrm{m\,s^{-1}} \le d_i \le 1000\ \mathrm{m\,s^{-1}}, \tag{15}$$

$$60\ \mathrm{s} \le t_i \le T - 60\ \mathrm{s}. \tag{16}$$

The burn times must be strictly increasing. A candidate with repeated, out-of-order or out-of-interval burn times is assigned a cost of ∞ before propagation. The 60 second margin applies at the beginning and end of the mission; it is not a minimum spacing between successive burns. No collision-avoidance constraint is included in these scoring functions.

Both scripts use a random seed of 42 and leave the remaining optimizer settings at the SciPy defaults. In the installed SciPy version 1.17.1, these include the `best1bin` strategy, a population size of $15(2N) = 30N$, and a maximum of 1000 generations. The search can stop earlier when the spread of population costs meets the default convergence tolerance of 0.01, with zero absolute tolerance. A final local polishing step is enabled by default. The returned sequence is therefore the best sequence found by this search, rather than a guaranteed global minimum. The previously defined final separation and velocity criteria must still be checked separately; they are not enforced as constraints by the optimizer.

3.4 Trajectory reconstruction and visualization

Once a sequence has been selected, I propagate it again to produce the trajectory plots. By default, the chaser is sampled at 1000 equally spaced output times per interval, and repeated interval endpoints are removed when joining the histories. These output times are used for plotting and do not force the ODE solver to use a fixed integration step. The target's continuous solution is evaluated at the chaser's sample times to calculate their separation.

For the 2 dimensional model, the plots show the trajectories around Earth and the separation over time, with the burn locations marked. For the 3 dimensional model, the plots include separation over time and both a view of the system around the Sun and a close-up of the rendezvous around Earth. For the Earth close-up, I subtract Earth's position from each spacecraft's position at the same time. This makes the motion around Earth easier to see without changing the calculated trajectories or burn directions.

4 Simulation Checks

The implementation was checked using 28 automated tests, all of which passed. These checks are described in Appendix B. The following checks focus on the numerical accuracy of the simulated motion and the reported rendezvous results.

4.1 Circular-orbit reference case

For the 2 dimensional model, I tested a circular orbit with a radius of 7.0×10^6 m and an initial speed of $\sqrt{\mu/r}$. I propagated the spacecraft for one orbital period, approximately 5828.52 s, using the setting defined for `solve_ivp` in the previous section. The simulated position was compared with the analytical circular motion at 1000 output times. The maximum position difference was 18.94 m, and the difference after one full orbit was 15.97 m. The maximum relative changes in specific orbital energy and angular momentum were 2.12×10^{-6} and 1.06×10^{-6}, respectively. This provides a reference check for the original Earth-only model.

4.2 Sensitivity to numerical tolerances

The solver calculates an approximation of the spacecraft motion. I therefore checked whether using lower numerical tolerances would significantly change the simulated results, particularly whether attempts previously classified as successful or unsuccessful would change classification. This check

was performed for the 3 dimensional model, but the same procedure can also be applied to the 2 dimensional model.

The validation experiments in Section 5 tested four sets of weights on eight retained initial-state cases, giving 32 burn sequences in the reported dataset. I simulated all 32 sequences again with tighter numerical tolerances, which require the solver to control its calculation error more closely. The burn times and magnitudes stayed the same; only the accuracy settings changed. The gravitational-body motion was also recalculated with the tighter settings. The settings are listed in Appendix A.

In the original simulations, two of the 32 retained attempts finished within 1000 m of the target with a relative speed no greater than 1 $\mathrm{m\,s^{-1}}$. The lower tolerances gave the same two passes and 30 failures. I then simulated the two passing sequences once more with a still lower relative tolerance, and both continued to pass.

The calculated distances did change, even though the success results did not. Table 1 shows both passing attempts from this final check, all using the overall-winning weights. For the 1 km initial offset, the final separation changed from approximately 16 m to 75 m. Both values are below the 1000 m limit defined earlier. Thus, lowering the tolerances changed some calculated final distances but did not change which of the tested attempts met the rendezvous criterion.

I also wanted to assess whether the additional accuracy would justify the extra computation required by lower tolerances. However, runtime was not measured, so the trade-off between accuracy and computation time was not quantified. For stricter rendezvous requirements, lower tolerances may be necessary to keep numerical errors small enough to assess success reliably.

Table 1: Original results and the final, tighter-tolerance check for the two successful attempts using the overall-winning weights. Burn times and magnitudes are unchanged.

Initial case	Original		Tighter check	
	e_r (m)	e_v (m/s)	e_r (m)	e_v (m/s)
1 km along-track	16.0	0.037	74.6	0.039
10 km along-track	149.5	0.251	172.8	0.243

5 Experiments and Results

5.1 Experiment setup and weight selection

For these experiments, I used the 3 dimensional model with the Sun fixed at the origin and Earth, the Moon and Jupiter moving under gravity. The mission duration was 3600 s, with two burns bounded between -1000 and 1000 $\mathrm{m\,s^{-1}}$ each. The saved initial body states, solver settings and optimizer seed of 42 were reused throughout the sweep and validation. Earlier experiments with different burn directions or magnitude limits were kept separate.

The initial state used to select the weights was

$$\begin{aligned} \text{Chaser: } & [7000000, -500000, 300000, 0, 7400, 500], \\ \text{Target: } & [7000000, 0, 0, 0, 7546, 0], \end{aligned}$$

where the entries are $[x, y, z, v_x, v_y, v_z]$ relative to Earth, in metres and metres per second. The initial search tested 1000 weight combinations, with each of the three weights taking one of 10 values from 0.001 to 1,000,000, increasing by a factor of 10 between values. Then 312 additional

refined combinations were tested within a smaller range. In the end, all 1312 trials returned finite results with valid burns and passed the sampled surface check. This status does not mean that they achieved rendezvous.

To compare results obtained with different weights, I used a similar dimensionless comparison score for every trial:

$$S = \left(\frac{e_r}{100\ \mathrm{m}}\right)^2 + \left(\frac{e_v}{0.1\ \mathrm{m\,s^{-1}}}\right)^2 + \frac{\Delta v_{\mathrm{total}}}{500\ \mathrm{m\,s^{-1}}}. \tag{17}$$

Here, e_r and e_v are the final separation and relative speed from each trial. The raw cost values from differently weighted objectives were not used to rank different weight tuples directly. This comparison score essentially makes all three values being measured for cost have the same impact.

I selected four weight tuples from the sweep: the lowest comparison score overall, the lowest comparison score from the first 1000 comparisons, the lowest final position error, and the lowest final velocity error. These weights were saved before testing the additional initial states. Table 2 lists the selected values and their subsequent validation results.

Table 2: Selected weights and project-criterion passes across the seven new initial states. The original selection case is excluded from the counts.

Selection	ω_r	ω_v	$\omega_{\Delta v}$	Passes
Overall (All 1312)	0.0075	5000	2.5	2/7
First 1000	0.001	10000	10	0/7
Position	1	0.1	1	0/7
Velocity	0.005	75000	2.5	0/7

5.2 Results for the original selection case

Table 3 compares the four selected weight tuples on the original initial states. None met the project rendezvous criterion. The overall-winning sequence applied a $+984.30\ \mathrm{m\,s^{-1}}$ burn at 61.67 s and a $-900.70\ \mathrm{m\,s^{-1}}$ burn at 354.19 s. Its minimum sampled separation was approximately 24.38 km during the trajectory, but it finished approximately 563.50 km from the target. Thus, its closest approach was not a terminal rendezvous.

The no-burn trajectory for this initial state entered Earth's sampled surface. Its final separation was approximately 2493.54 km and its relative speed was $2114.99\ \mathrm{m\,s^{-1}}$, but these terminal values come from continuing the point-mass simulation after surface intersection. They do not describe a physically flyable coast trajectory.

Table 3: Results for the original weight-selection experiment.

Selection	Final separation (km)	Relative speed (m/s)	Total Δv (m/s)
Overall (1312)	563.50	134.36	1885.00
First 1000	563.77	133.95	1898.36
Position	563.39	390.53	1222.83
Velocity	564.49	133.90	1662.79

5.3 Validation on different initial states

The validation analysis includes the four selected weight tuples on the original case and seven new cases, giving 32 powered optimization trials. For each of the eight retained initial-state cases, I also simulated the spacecraft motion without any burns to create a baseline to compare the final results to. For each combination of weights and initial states, I searched for new burn times and magnitudes.

The new cases included along-track offsets of 1 km and 10 km, a radial offset, a cross-track offset, a mixed position and velocity offset, a higher target orbit, and an inclined target orbit. The full initial states, weights, selected burns and measurements are recorded in Appendix C. Table 4 shows the production results for the overall-winning weights, and Figure 2 compares their final errors with the no-burn controls.

Table 4: Production validation results using the overall-winning weights. A pass requires both $e_r \leq 1000$ m and $e_v \leq 1$ m/s.

Case	e_r (m)	e_v (m/s)	Total Δv (m/s)	Pass
1 km along-track	16.0	0.037	6.30	Yes
10 km along-track	149.5	0.251	101.50	Yes
10 km radial	2871.1	5.224	372.01	No
10 km cross-track	13639.8	0.340	18.89	No
Mixed, about 100 km	41209.7	3.272	151.45	No
Higher target orbit	2086.6	1.972	34.99	No
Inclined target orbit	6166.4	1.216	758.70	No

Figure 1 shows the two successful attempts using the overall-winning weights. The orbit views show their motion around Earth, while the close-ups show the chaser's position relative to the target. The paths nearly overlap at the scale of the orbit, so the close-ups make their separation easier to see.

The overall-winning weights also achieved rendezvous in the 1 km and 10 km along-track cases, whereas both no-burn baselines failed. For the 1 km case, the optimized burns reduced the final separation from 2568.4 m to 16.0 m and the relative speed from 2.011 $\mathrm{m\,s^{-1}}$ to 0.037 $\mathrm{m\,s^{-1}}$. For the 10 km case, they reduced the final separation from 91560.0 m to 149.5 m and the relative speed from 83.754 $\mathrm{m\,s^{-1}}$ to 0.251 $\mathrm{m\,s^{-1}}$.

For the successful 1 km case, the selected burns were $+3.127\ \mathrm{m\,s^{-1}}$ at 60.014 s and $-3.173\ \mathrm{m\,s^{-1}}$ at 216.811 s. Figure 3 shows its separation and relative speed over the mission. The lower-tolerance recheck in Table 1 also met the secondary, stricter criterion of 100 m and 0.1 $\mathrm{m\,s^{-1}}$.

Across the seven new states, the best weights from the initial 1000 combinations, the position selection and the velocity selection all failed to meet the rendezvous criterion. As an example of the difference between matching position and achieving rendezvous, the position selection finished the 1 km case with a reported separation of 0.188 m but a relative speed of 11.829 $\mathrm{m\,s^{-1}}$. It therefore failed the velocity criterion. All measurements in this section are from the saved production runs; the additional numerical rechecks are reported separately in Section 4. The validation used one seed and seven selected new states, so the pass counts describe these experiments rather than a general mission success probability.

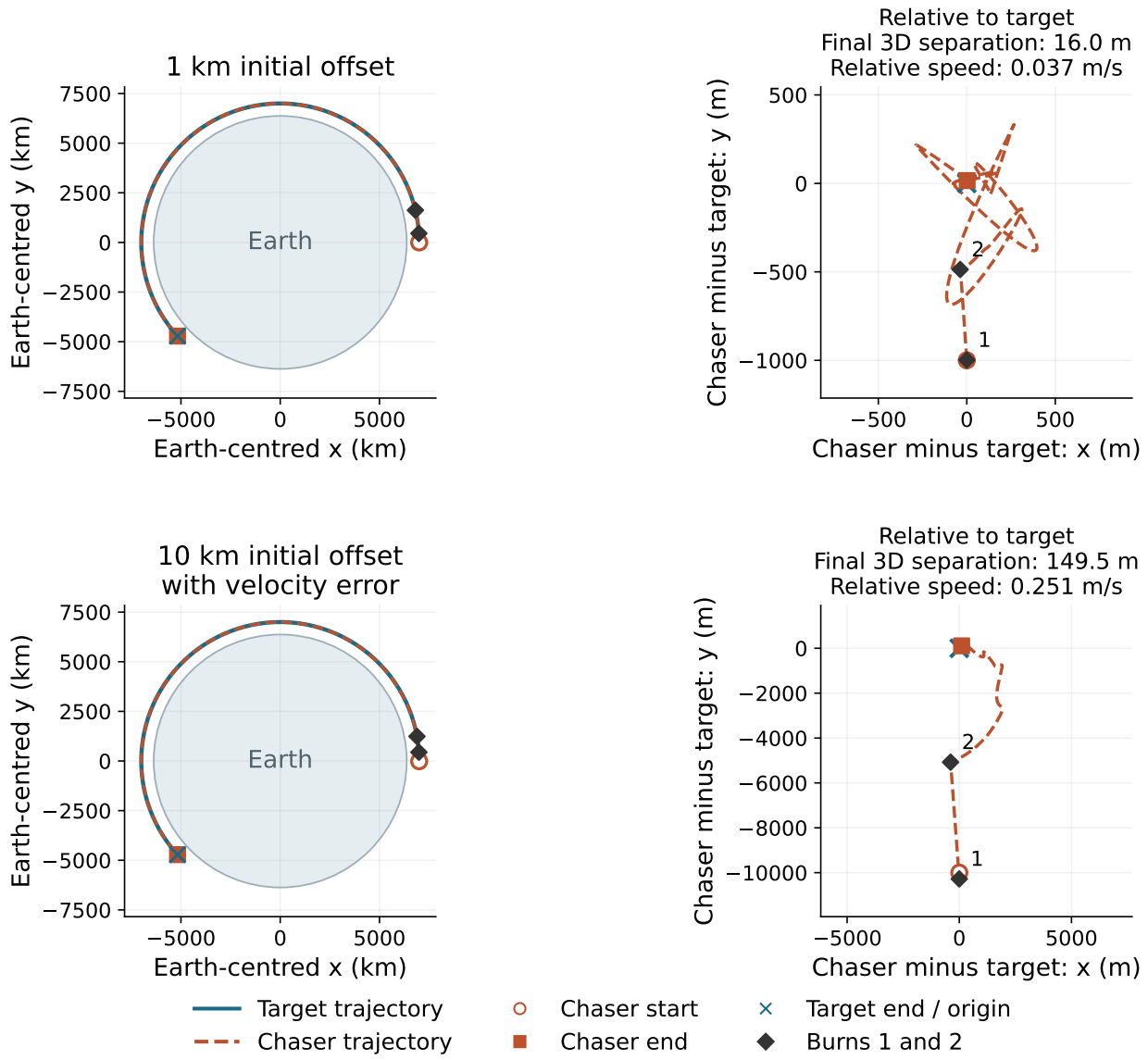


Figure 1: Successful validation attempts using the overall-winning weights. Each row shows an Earth-centred orbit view (left) and the chaser's motion relative to the target (right). Both views are x–y projections of the 3 dimensional trajectories; the final separation and relative speed use all three components. Numbered diamonds in the close-ups mark the two burns. The target is at the origin in each close-up.

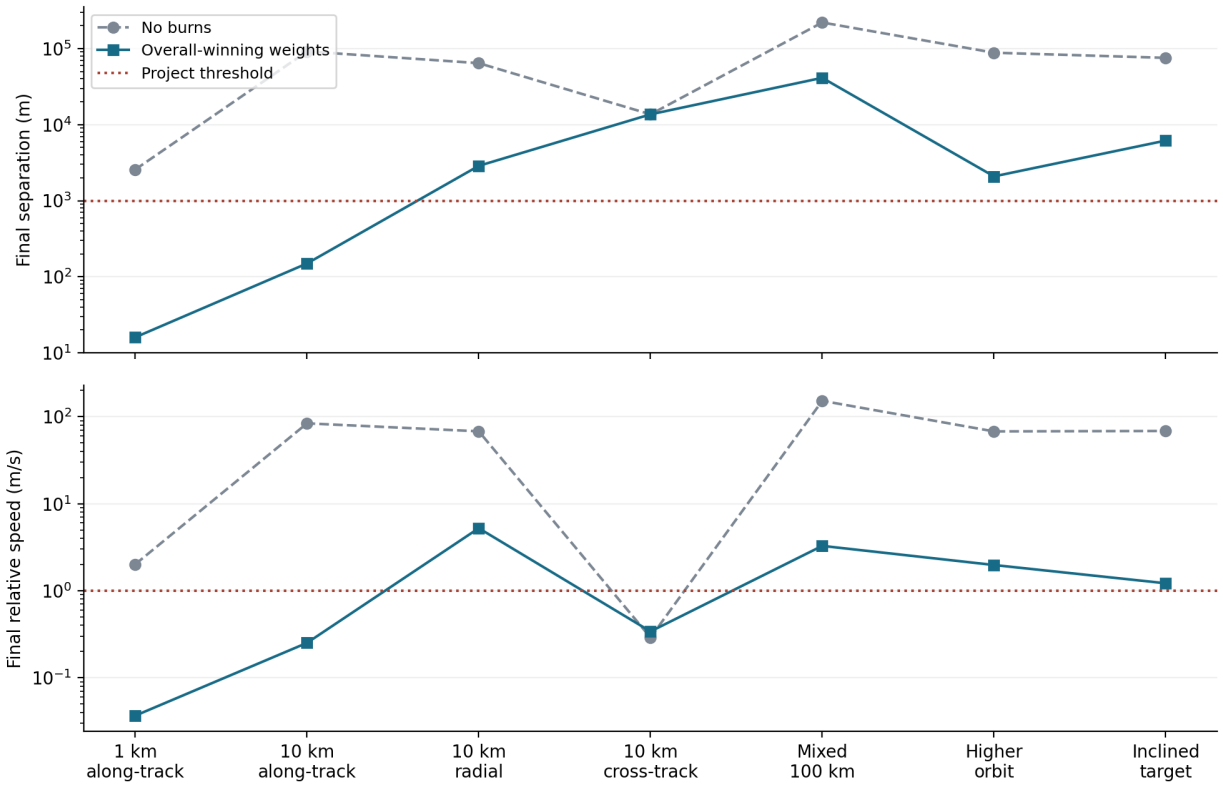


Figure 2: Final separation and relative speed for the seven new cases, comparing the overall-winning weights with no burns. The vertical scales are logarithmic; dotted lines show the project thresholds.

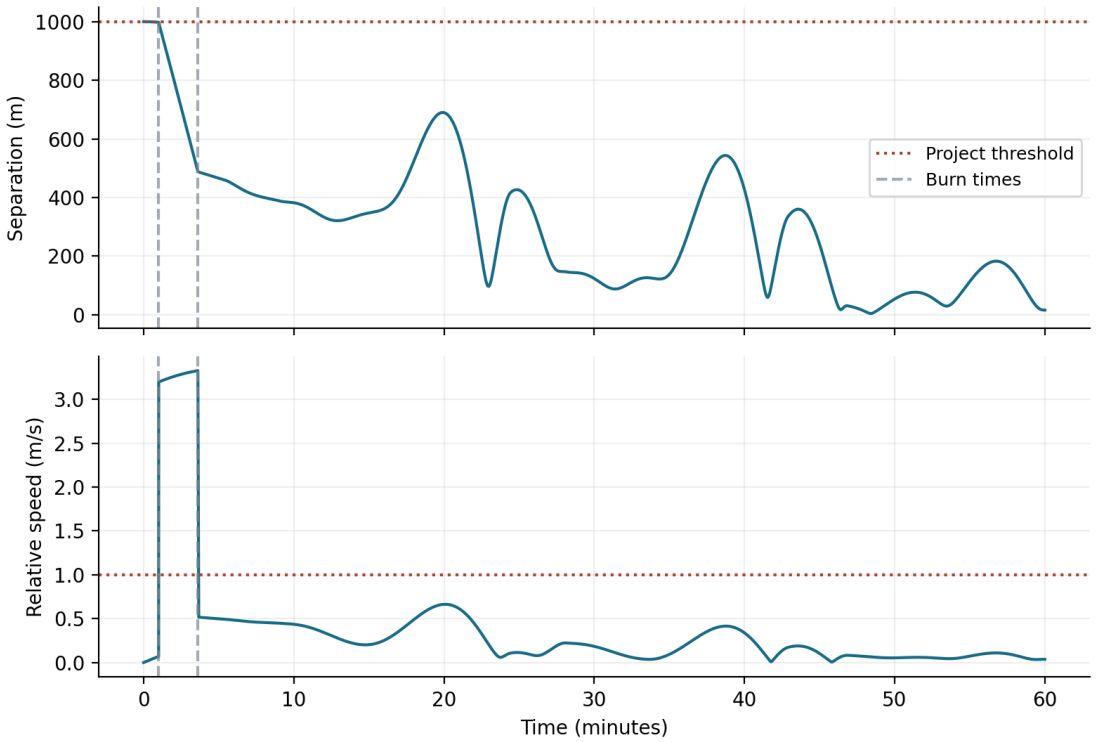


Figure 3: Production separation and relative speed for the 1 km along-track case using the overall-winning weights. Dashed lines mark the two burns; dotted lines show the project success thresholds.

6 Discussion

Writing prompt: Explain what the results mean, connecting trajectory changes to the physics rather than simply describing plots.

- Why did the chosen burn sequence improve the rendezvous attempt?
- How did timing and velocity changes affect relative motion?
- What trade-offs are supported by the experiments?
- Were remaining errors associated with numerical accuracy, search limits, burn restrictions, or objective weighting? Distinguish evidence from possible explanations.
- What do the simplified dynamics omit, and how does that limit conclusions about a real mission?

Writing prompt: Use "best solution found within the tested search" unless you have evidence of global optimality. Avoid general conclusions about burn count or fuel savings from a single case.

7 Conclusion and Future Work

Writing prompt: Answer the central question directly. Summarize the strongest quantitative finding, the main physical insight, and the principal limitation. Suggest one or two specific next steps justified by the results.

References

Writing prompt: Cite the project brief, any external theory used, numerical-method documentation, and any adapted code. Add bibliographic entries as you write.

A Reproducibility Details

Writing prompt: Record the script/version used, software dependencies, exact initial states, solver and optimizer settings, seeds, objective weights, and final burn sequence. Include supporting checks or additional plots here when they would interrupt the main argument.

A.1 Settings for the numerical accuracy checks

Table 5 lists the solver tolerances used for the 3 dimensional accuracy checks in Section 4.2. Each repeated simulation used the same initial states, burn times and burn magnitudes as the original result. Both the gravitational-body trajectories and spacecraft trajectories were recalculated; no new optimization was run.

Table 5: Solver settings for the original 3 dimensional simulations and their numerical accuracy checks.

Calculation	Relative tolerance	Absolute tolerance	Sequences
Original simulations	10^{-6}	10^{-6}	32
First accuracy check	10^{-10}	10^{-8}	32
Final accuracy check	10^{-12}	10^{-8}	2

The final check was restricted to the two retained sequences that passed the project criterion. Between the first and final accuracy checks, the largest change in final separation for these sequences was less than 0.004 m, and the largest change in relative speed was less than $5 \times 10^{-6}\ \mathrm{m\,s^{-1}}$. Both passed the criterion in both checks. This comparison concerns numerical propagation of fixed burn sequences, rather than the reliability of the optimizer.

B Implementation Checks

To check that the program behaves as intended, I used 28 automated tests covering the body simulation, burns, trajectory plots, weight sweep and validation reporting. All 28 tests passed. Table 6 shows the five test groups. Individual tests can check several inputs.

Table 6: Automated implementation checks. All listed test groups passed.

Part of the program	Number of tests
Gravitational-body simulation	5
Earth-relative burns and invalid burn times	2
Trajectory calculation and visualization	4
Weight sweep and saved-run handling	13
Validation scenarios, selection and reporting	4
Total	28

B.1 Gravitational-body simulation

For a prescribed set of initial body states, I compare Earth's calculated acceleration with a separately evaluated sum of the gravitational contributions from the Sun, Moon and Jupiter. I also check that the position derivatives equal the supplied velocities and that all six derivatives of the Sun's state are zero.

A separate one-day propagation uses 100 output times. This checks that the initial state is preserved at the first output, all state values remain finite, Earth moves during the simulation, and the Sun's position and velocity remain exactly zero at every output. The relative difference between the initial and final total energy must be less than 10^{-6}. This energy includes the moving bodies' kinetic energy and the gravitational potential energy of the model. Total momentum conservation is not required, since the Sun is physically held fixed.

The initial-state reader is tested using controlled sample responses in the format returned by JPL Horizons. The six position and velocity entries must be extracted correctly and multiplied by 1000 to convert to SI units. A response with no state vector must be rejected. These checks test the reader, not the accuracy of the Horizons data. Invalid durations, state-array dimensions, output sample counts, nonzero initial Sun states and coincident bodies are also tested.

B.2 Burn direction and application

To isolate the burn calculation, I use a test case with no gravitational acceleration. The chaser moves at constant velocity between burns, allowing its final position and velocity to be calculated directly. A burn is applied at 300 s during a 600 s trajectory. The tested signed magnitudes are $+100$, -100 and $0\ \mathrm{m\,s^{-1}}$.

Earth's prescribed velocity changes with time in this test. This checks that the burn uses Earth's velocity at the actual burn time, rather than its initial velocity. The same constant velocity is also added to both Earth and the chaser in a second version of each case. Their relative motion is unchanged, so the applied velocity change must remain the same.

For each case, the calculated final state is compared with the direct solution using an absolute tolerance of 10^{-5} in the SI state components. Both the trajectory reconstruction and candidate

scoring are also compared with the expected cost to six decimal places. This checks the direction, sign, magnitude and timing of the impulse, as well as consistency between the two calculation paths.

For both the 2 dimensional and 3 dimensional models, burns outside the simulation interval, burns at repeated times, and burns in the wrong order must return an infinite cost before any trajectory is propagated. The sweep's separate output check accepts magnitudes exactly at $\pm 1000\ \mathrm{m\,s^{-1}}$ and rejects a magnitude of $1000.001\ \mathrm{m\,s^{-1}}$, nonfinite values and unordered times.

B.3 Trajectory calculation and visualization

For a short 3 dimensional trajectory, I compare the cost calculated during scoring with the cost obtained when reconstructing the same burn sequence for plotting. The values must agree to seven decimal places in the test case. The plotted separation is checked against the distance calculated directly from the spacecraft states at matching times, including differences in the z direction.

The plots are checked for correct positions relative to the Sun and Earth, equal scales along the three axes, and use of the specified body radii. The figures are rendered to check the complete plotting operation.

Additional tests check that the optimizer receives the precomputed body trajectories, that the interactive program accepts six-component spacecraft states, and that the displayed final and minimum sampled separations match the calculated values. A shortened optimization is used for the automated integration checks only; the production sweep and validation use the full optimizer settings described in the report.

B.4 Weight sweep and preservation of results

The sweep tests check the initial 1000 weight combinations, the refinement values and the exclusion of duplicate weight combinations. A controlled test also checks that refinement is centred on the best result from the initial search. The fixed comparison score is checked using values with a directly calculable result, and failed or surface-intersecting trials are excluded from selection.

To check reproducibility, the tests verify that a new sweep can reuse saved initial body states without another network request. Changes to the gravitational-body code must change the recorded model identifier, and a saved run with a different identifier must be rejected before further trials are started. A dry run must make neither a network request nor an optimizer call.

The saved-run tests check that completed trials survive an interruption, failed trials are recorded, and a requested trial limit stops further work. Resuming must preserve the existing configuration and CSV rows, skip completed combinations, and keep the refinement centred on the same initial-search result. A CSV with an incomplete final line must be rejected without modifying the saved file.

B.5 Validation selection and classification

The validation tests check that the eight scenarios are distinct, contain finite six-component states, and include changes to the target orbit. Candidate weights must be selected only from the saved source-sweep measurements, before the validation cases are used.

For the stricter diagnostic criterion, test cases verify that both the final position and velocity limits must be met and that a surface-intersecting or failed trial cannot be counted as a pass. Reporting is also checked with a deliberately failed trial, so that failure information is retained, an incomplete batch remains labelled incomplete, and the original selection case is excluded from the new-state summary.

C Full Validation Data

This appendix records the eight retained initial-state cases, four selected weight tuples, 32 optimized burn sequences and eight no-burn controls used in Section 5. An initial proximity case that already met the success criterion without burns is excluded from the reported dataset. The unfiltered experiment records are archived separately. These are the original validation results, using relative and absolute solver tolerances of 10^{-6}. The tighter-tolerance checks are reported separately in Section 4 and Appendix A. Values are rounded for display; the full-precision records are saved in the configuration and results JSON files under `outputs/weight_validation_fixed_sun_20261029_144240/`.

C.1 Shared settings and selected weights

Every case uses the fixed-Sun model with Earth, Moon and Jupiter moving under gravity. The initial body states correspond to 27 October 2026 at 12:00:00 UTC. The mission lasts 3600 s, with two burns bounded between -1000 and 1000 $\mathrm{m\,s^{-1}}$ each. Burn times lie between 60 and 3540 s and must be strictly increasing. The optimizer seed is 42, with the full differential-evolution settings described in Section 3. Powered trajectories use 1000 samples per interval; no-burn controls use 3000 output samples.

Selection	ω_r	ω_v	$\omega_{\Delta v}$
Overall	0.0075	5000	2.5
First 1000	0.001	10000	10
Position	1	0.1	1
Velocity	0.005	75000	2.5

In the case tables, a project pass requires both a final separation no greater than 1000 m and a relative speed no greater than 1 $\mathrm{m\,s^{-1}}$. It also requires a trajectory that passes the sampled surface check. The clearance is the minimum sampled altitude of either spacecraft above any of the four modelled body surfaces. A positive clearance does not rule out a surface crossing between samples. The closest-approach time refers to the minimum sampled chaser–target separation, not to surface clearance. The score S uses the fixed comparison scales from Section 5; the objective J uses the selected weights and should not be compared directly between different weight tuples. Elapsed times are the recorded powered-trial times and are not timing measurements of the tighter-tolerance accuracy checks.

C.2 Initial gravitational-body states

The following positions and velocities initialize the body propagation, with the Sun fixed at the origin. Displayed components are rounded to six decimal places. Positions are in metres and velocities in metres per second. The same saved values are used for every validation case.

Body / quantity	x or v_x	y or v_y	z or v_z
Sun position	0	0	0
Sun velocity	0	0	0
Earth position	123676106836.450897	82518717047.766449	-6197900.438908
Earth velocity	-17006.728051	24659.141135	-1.526054
Moon position	123901995387.667694	82804221362.851425	25736913.855627
Moon velocity	-17864.710185	25312.11686	5.584688
Jupiter position	-544696242047.874573	579247541988.982056	9780652767.300129
Jupiter velocity	-9677.775054	-8349.347558	251.206352

C.3 Original weight-selection case

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [7000000,\ -500000,\ 300000,\ 0,\ 7400,\ 500]$$
$$\mathbf{s}_{t,0} = [7000000,\ 0,\ 0,\ 0,\ 7546,\ 0].$$

The no-burn control enters a modelled body surface. Its terminal measurements describe continued point-mass motion after that intersection and are not a physically flyable coast result.

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	2493544.031	2114.990	0.000	No
Overall	563497.512	134.362	1885.000	No
First 1000	563768.674	133.947	1898.361	No
Position	563393.398	390.532	1222.828	No
Velocity	564486.810	133.905	1662.791	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	583553.944	1.200	-64.134	Intersection
Overall	24378.862	2345.864	536.917	Clear
First 1000	29397.714	2346.029	537.056	Clear
Position	404349.762	2711.820	255.257	Clear
Velocity	27007.816	2345.162	526.810	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	984.301	61.666	-900.698	354.193
First 1000	990.973	62.325	-907.389	354.618
Position	594.733	501.987	-628.095	1182.311
Velocity	871.807	62.157	-790.984	402.084

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	2.47174×10^{9}	2.47174×10^{9}	3.35583×10^{7}	21.851
First 1000	4.97271×10^{8}	4.97271×10^{8}	3.35777×10^{7}	16.822
Position	3.17412×10^{11}	3.17412×10^{11}	4.69928×10^{7}	14.423
Velocity	2.93802×10^{9}	2.93802×10^{9}	3.36576×10^{7}	16.844

C.4 1 km along-track offset

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [7000000,\ -1000,\ 0,\ 0,\ 7546,\ 0]$$
$$\mathbf{s}_{t,0} = [7000000,\ 0,\ 0,\ 0,\ 7546,\ 0].$$

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	2568.399	2.011	0.000	No
Overall	16.013	0.037	6.300	Yes
First 1000	1339.365	0.033	1.237	No
Position	0.188	11.829	10.853	No
Velocity	1342.175	0.021	1.250	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	815.988	782.661	627.884	Clear
Overall	4.232	2905.752	628.577	Clear
First 1000	657.193	798.445	627.972	Clear
Position	0.188	3600.000	627.884	Clear
Velocity	657.068	795.448	627.971	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	3.127	60.014	-3.173	216.811
First 1000	0.147	60.134	1.090	3539.247
Position	-10.839	3360.624	0.014	3502.942
Velocity	0.147	60.739	1.103	3539.288

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	24.3693	24.3693	0.172144	62.003
First 1000	1817.29	1817.29	179.503	26.756
Position	24.8811	24.8811	13992.7	28.417
Velocity	9044.55	9044.55	180.192	24.569

C.5 10 km along-track offset with velocity error

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [7000000,\ -10000,\ 0,\ 0,\ 7541,\ 0]$$
$$\mathbf{s}_{t,0} = [7000000,\ 0,\ 0,\ 0,\ 7546,\ 0].$$

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	91559.958	83.754	0.000	No
Overall	149.498	0.251	101.496	Yes
First 1000	14246.394	0.357	18.225	No
Position	0.915	129.888	127.854	No
Velocity	14255.823	0.338	18.115	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	10005.994	1.200	605.939	Clear
Overall	149.498	3600.000	627.382	Clear
First 1000	7439.083	801.850	620.729	Clear
Position	0.915	3600.000	607.382	Clear
Velocity	7441.313	801.822	620.726	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	53.039	60.003	-48.457	165.889
First 1000	6.556	60.141	11.669	3538.862
Position	16.694	1658.268	-111.160	3034.338
Velocity	6.553	60.135	11.562	3538.751

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	736.622	736.622	8.74313	32.013
First 1000	204420	204420	20308.8	18.570
Position	1815.77	1815.77	1.68708×10^6	179.335
Velocity	1.02477×10^6	1.02477×10^6	20334.3	20.823

C.6 10 km radial offset with velocity error

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [7010000,\ 0,\ 0,\ 0,\ 7541,\ 0]$$
$$\mathbf{s}_{t,0} = [7000000,\ 0,\ 0,\ 0,\ 7546,\ 0].$$

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	64244.593	67.886	0.000	No
Overall	2871.060	5.224	372.006	No
First 1000	6424.565	0.231	12.672	No
Position	1.352	83.899	78.644	No
Velocity	6446.499	0.181	12.709	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	10000.019	1.200	628.713	Clear
Overall	2795.499	3472.917	623.930	Clear
First 1000	6424.565	3600.000	621.381	Clear
Position	1.352	3600.000	617.211	Clear
Velocity	6446.499	3600.000	621.409	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	-188.094	62.864	183.912	73.437
First 1000	-5.123	60.070	-7.549	3312.871
Position	-11.520	1582.937	67.123	3037.038
Velocity	-5.116	60.144	-7.594	3298.626

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	199226	199226	3554.51	23.832
First 1000	41936.9	41936.9	4132.88	20.372
Position	784.37	784.37	703900	197.872
Velocity	210288	210288	4159.05	23.737

C.7 10 km cross-track offset

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [7000000,\ 0,\ 10000,\ 0,\ 7546,\ 10]$$
$$\mathbf{s}_{t,0} = [7000000,\ 0,\ 0,\ 0,\ 7546,\ 0].$$

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	13642.328	0.289	0.000	No
Overall	13639.754	0.340	18.890	No
First 1000	13644.654	0.128	2.598	No
Position	13640.143	5.225	338.290	No
Velocity	13640.070	0.130	2.477	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	75.660	2151.117	628.713	Clear
Overall	75.644	2150.034	628.713	Clear
First 1000	75.257	2150.724	628.713	Clear
Position	76.220	2151.397	628.713	Clear
Velocity	56.514	2151.593	628.713	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	9.324	2705.143	-9.567	2726.299
First 1000	1.329	3060.646	-1.269	3173.905
Position	-166.641	3529.139	171.649	3532.811
Velocity	1.252	1695.164	-1.225	1711.021

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	1.39595×10^6	1.39595×10^6	18615.9	11.465
First 1000	186368	186368	18619.3	16.892
Position	1.86054×10^8	1.86054×10^8	21335.6	10.267
Velocity	931536	931536	18606.9	15.163

C.8 Mixed position and velocity offset

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [7020000,\ -100000,\ 30000,\ 10,\ 7526,\ 30]$$
$$\mathbf{s}_{t,0} = [7000000,\ 0,\ 0,\ 0,\ 7546,\ 0].$$

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	219999.841	152.028	0.000	No
Overall	41209.672	3.272	151.451	No
First 1000	122184.227	5.458	93.965	No
Position	41171.616	942.430	984.541	No
Velocity	123129.685	3.085	94.292	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	85783.428	1130.777	541.613	Clear
Overall	13288.730	2243.269	611.466	Clear
First 1000	73825.987	1223.691	562.299	Clear
Position	41171.616	3600.000	541.613	Clear
Velocity	74039.037	1220.587	561.993	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	72.086	81.404	-79.365	659.926
First 1000	11.273	60.503	82.692	3539.619
Position	59.019	3118.131	-925.522	3334.474
Velocity	11.092	61.030	83.200	3539.701

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	1.27907×10^{7}	1.27907×10^{7}	170895	21.329
First 1000	1.52278×10^{7}	1.52278×10^{7}	1.49588×10^{6}	14.033
Position	1.69519×10^{9}	1.69519×10^{9}	8.8987×10^{7}	15.154
Velocity	7.65188×10^{7}	7.65188×10^{7}	1.51704×10^{6}	15.981

C.9 Higher target orbit

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [8000000,\ -10000,\ 2000,\ 0,\ 7053.682597,\ 2]$$
$$\mathbf{s}_{t,0} = [8000000,\ 0,\ 0,\ 0,\ 7058.682597,\ 0].$$

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	88090.624	67.723	0.000	No
Overall	2086.614	1.972	34.991	No
First 1000	2163.646	1.941	39.662	No
Position	2080.819	721.357	1221.156	No
Velocity	2126.342	1.939	155.290	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	10204.390	1.200	1602.625	Clear
Overall	472.032	2670.307	1626.833	Clear
First 1000	823.314	2670.765	1627.040	Clear
Position	2080.819	3600.000	1602.636	Clear
Velocity	851.792	2619.307	1627.360	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	19.744	61.066	-15.247	419.304
First 1000	22.059	61.189	-17.603	376.716
Position	258.093	3067.134	-963.063	3383.932
Velocity	79.937	135.170	-75.353	209.993

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	52182.1	52182.1	824.264	30.301
First 1000	42740.3	42740.3	844.838	29.739
Position	4.38307×10^6	4.38307×10^6	5.2036×10^7	17.870
Velocity	305063	305063	828.534	23.245

C.10 Target orbit inclined by 20 degrees

The initial spacecraft states are relative to Earth, in the order $[x, y, z, v_x, v_y, v_z]$, with positions in metres and velocities in $\mathrm{m\,s^{-1}}$:

$$\mathbf{s}_{c,0} = [7000000,\ -10000,\ 2000,\ 0,\ 7085.966663,\ 2582.900798]$$
$$\mathbf{s}_{t,0} = [7000000,\ 0,\ 0,\ 0,\ 7090.966663,\ 2580.900798].$$

Terminal results and total velocity change.

Selection	e_r (m)	e_v (m/s)	Total Δv (m/s)	Project pass
No burns	75387.384	68.470	0.000	No
Overall	6166.398	1.216	758.701	No
First 1000	6271.174	1.211	32.050	No
Position	6163.915	417.146	843.021	No
Velocity	13868.670	1.211	15.416	No

Closest approach and sampled surface clearance.

Selection	Min. separation (m)	Time (s)	Clearance (km)	Surface check
No burns	10204.387	1.200	610.146	Clear
Overall	888.819	1998.606	627.741	Clear
First 1000	132.910	1981.222	626.807	Clear
Position	6163.915	3600.000	610.146	Clear
Velocity	8793.502	888.545	621.810	Clear

Selected burns. Signed magnitudes are in m/s and times are in seconds.

Selection	d_1	t_1	d_2	t_2
Overall	381.165	117.033	-377.536	129.733
First 1000	17.714	64.014	-14.336	410.337
Position	225.679	3079.702	-617.342	3308.088
Velocity	5.344	60.499	10.073	3536.204

Recorded costs and computation times.

Selection	Optimizer J	Reconstructed J	Score S	Elapsed (s)
Overall	294473	294473	3951.81	21.014
First 1000	54314.5	54314.5	4079.49	26.231
Position	3.80121×10^7	3.80121×10^7	1.74049×10^7	26.532
Velocity	1.07172×10^6	1.07172×10^6	19380.7	18.564